SUPERVISED MACHINE LEARNING CLASSIFICATION PROJECT

Student Assignment

Dataset

Breast Cancer Wisconsin (Original)

Dataset link: UCI Machine Learning Repository — Breast Cancer Wisconsin (Original)

Objective

Develop a supervised machine learning classification system using the provided dataset. Analyze and preprocess the data, train multiple classification models, and identify the model that achieves the best overall performance.

Required Tasks

Task 1: Dataset Exploration

- Load the dataset using Pandas.
- Explore its structure, dimensions, column names, and data types.
- Identify the target variable and predictive features.
- Investigate feature meanings and target labels using the dataset documentation.
- Analyze descriptive statistics, missing values, duplicate records, and class distribution.

Task 2: Data Preprocessing

- Handle missing values using an appropriate technique.
- Check for and handle duplicate records.
- Remove irrelevant features, such as unique identifiers.
- Split the dataset into training and testing sets using an 80:20 ratio.
- Apply standardization where necessary.
- Ensure preprocessing is performed without data leakage.

Required Tasks (Continued)

Task 3: Exploratory Data Analysis (EDA)

- Create histograms to explore feature distributions.
- Create a count plot to visualize the target class distribution.
- Use boxplots to compare feature distributions and identify potential outliers.
- Create a correlation heatmap to investigate relationships among features.
- Visualize missing values using counts or percentages.
- Create before-and-after standardization plots.
- Give each visualization a title, labeled axes, and a brief interpretation.

Task 4: Model Implementation

Implement and train each of the following supervised machine learning algorithms:

- K-Nearest Neighbors (KNN)
- Decision Tree (DT)
- Random Forest (RF)
- Artificial Neural Network (ANN)
- Support Vector Machine (SVM)

Use appropriate libraries, such as Scikit-learn, and perform hyperparameter tuning using cross-validation or a suitable validation procedure.

Task 5: Model Evaluation

Evaluate every model using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC (Area Under the Receiver Operating Characteristic Curve)

Additional requirements:

- Generate a confusion matrix for each model.
- Present all results in one comparison table.
- Treat malignant cases as the positive class.
- Explain the evaluation metrics and interpret the results.
- Compare the models and justify which one performs best overall.

Task 6: Presentation Slides

Prepare a slide presentation explaining your complete project. Include the following:

- Problem statement and project objective.
- Dataset source and exploration findings.
- Data cleaning and preprocessing techniques.
- EDA charts and their interpretations.
- Explanation of the five implemented algorithms.
- Evaluation metrics and confusion matrices.
- Comparison table of all models.
- Best-performing model and justification for its selection.
- Conclusion and possible future improvements.

Submission Requirements

- A Jupyter Notebook (.ipynb) containing your code, outputs, charts, and explanations.
- Presentation slides (.pptx or .pdf) explaining the complete project.

Important Notes

- Use experimental results to select the best model; do not assume that one algorithm will outperform the others.
- Fit preprocessing transformations on the training data only.
- Keep the testing data separate during model selection and hyperparameter tuning.